

A Survey of Rural and Urban Road Identification via Satellite Imagery

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Abstract—The automatic identification of road networks from satellite and aerial imagery has become a critical task for many applications, such as transportation planning, particularly in regions where cartographic data are incomplete or outdated. This paper presents a comprehensive survey of road extraction methods applied to both urban and rural environments, covering research published over the past decade. Modern data-driven techniques are reviewed. Particular emphasis is placed on convolutional neural networks (CNN), fully convolutional networks (FCN), encoder-decoder architectures (DCED), and generative adversarial networks (GAN), highlighting their respective strengths and limitations under varying environmental conditions. In addition, commonly used satellite sensors, annotation sources, benchmark datasets, and evaluation metrics are discussed to provide a unified perspective on experimental practices in the literature. By comparing reported results across a wide range of methods, this survey identifies prevailing trends in model performance and exposes challenges related to inconsistent evaluation protocols and dataset variability. The paper concludes by emphasizing the need for standardized benchmarks and evaluation frameworks to enable more reliable and meaningful comparisons between road extraction approaches.

Index Terms—Road extraction, satellite imagery, remote sensing, convolutional neural networks, deep learning, urban and rural road detection, semantic segmentation, benchmark datasets, evaluation metrics

I. INTRODUCTION

Accurate and up-to-date road maps are crucial for transportation planning, emergency response, and environmental monitoring. However, many regions, particularly in remote or developing areas, remain poorly mapped, as traditional mapping efforts are often slow, inconsistent, and difficult to scale.

Recent advances in artificial intelligence, especially deep learning, applied to high-resolution satellite imagery, have greatly enhanced the feasibility of automated road extraction, with implementations now effective for road detection in both urban and rural environments.

This paper surveys and analyzes 20 studies published over the last decade, with the aim of identifying trends and highlighting which architectures show the most promise for road extraction tasks.

The paper is organized into six sections: the current introduction, methods for data gathering and annotation, heuristic

approaches, data-driven methods, reported results, evaluation metrics, and conclusions.

II. DATA GATHERING

Accurate road extraction depends heavily on the quality, diversity, and resolution of the imagery used for training and evaluation. Modern approaches typically rely on a combination of publicly available satellite data, commercial high-resolution sensors, and crowdsourced annotations. This section summarizes the most commonly used sources and benchmark datasets in the literature. Each of the references following a list item is representative of a paper using said data collection.

A. Satellite and Aerial Imagery Sources

High-resolution satellite platforms provide detailed spectral and spatial information suitable for distinguishing between road surfaces, vegetation, and surrounding terrain. Commonly used satellites include:

- **IKONOS**: provides 1 m panchromatic and 4 m multispectral resolution imagery. [1]
- **QuickBird**: offers black and white 61 centimeter resolution and 2.44-1.63 meter multispectral resolution imagery, used mostly for urban road extraction studies. (reentered Earth after orbit decay in 2015 [2] [3]) [1] [4]
- **Pleiades-1A/1B**: delivers 0.5 m spatial resolution, particularly useful for modeling complex urban road networks [5].
- **GeoEye**: another high-resolution commercial sensor often used for cross-sensor generalization experiments [5].

In addition to satellite data, very-high-resolution imagery from **unmanned aerial vehicles (UAVs)** has gained popularity. UAV imagery typically ranges from a few centimeters to tens of centimeters per pixel and allows precise extraction of small rural or unpaved road segments [6].

B. Freely Accessible Annotation Sources

Reliable road annotations are essential for supervised learning. Two widely used sources are:

- **OpenStreetMap (OSM)** A globally crowdsourced mapping platform frequently used to generate or refine road labels for training and benchmarking [7] [4].

- **Google Earth** Provides multi-source RGB aerial and satellite images with varying resolutions. These images are often used for visual comparison and manual annotation [8] [4] [9].

C. Benchmark Datasets

Several curated datasets have become standard benchmarks for evaluating road extraction models:

- **Massachusetts Roads Dataset:** Contains 1711 RGB images with 1 m spatial resolution at a size of 1500 x 1500 pixels, covering an area of 2.25 square kilometers. It is widely used for evaluating CNN-based and FCN-based approaches [10] [5] [11] [12] [13] [14] [15] [16].
- **UAV-based Road Datasets:** Provide extremely high-resolution imagery for fine-grained road segmentation tasks, especially in rural or forested areas [6].
- **TerraSAR-X Road Dataset:** Used to test the robustness of algorithms under SAR imaging conditions, where optical cues are absent [17].
- **Abu Dhabi Dataset:** The size of each image in this dataset is of 1500 x 1500 pixels composed of RGB channels and a spatial resolution of 0.5 m per pixel. It is composed of images of roads, buildings and background of Abu Dhabi [12].
- **Conghua Dataset:** The dataset image resolution is of 0.2 m per pixel, which consists of RGB images. There are 47 aerial images in this dataset, and each image consists of 3 x 6000 x 6000 pixels. [16]
- **European Road Dataset:** Composed of 200 1600x1550 RGB training images (aprox. $276.4km^2$) for training with a spatial resolution is about $1m^2$ per pixel. [18]
- **Mação Dataset:** Made of sixty-one high-resolution orthorectified aerial images consisting of RGBI channels with a spatial resolution of 0.25 m. Each image presenting 16,000 x 10,000 pixels covering an area of 4 km by 2.5 km. [19]

These datasets collectively cover diverse environmental conditions from dense urban grids to remote rural and tropical regions supporting the development of generalizable road extraction models.

III. HEURISTIC METHODS

Early road extraction pipelines relied on handcrafted features extraction such as texture analysis, morphological filtering, and edge detection. Although these methods are computationally efficient, they generally struggle when applied to complex or heterogeneous terrain, or when image obstructions are present, occluding road features. Consequently, heuristic-based approaches have seen a significant decline in usage in recent years. In this survey, only one representative study on heuristic methods is included, reflecting the authors' assessment that such approaches are largely obsolete and rarely adopted in contemporary road extraction research.

Texture-based progressive analysis and mathematical morphology techniques have been widely applied; however, they often fail to generalize across different road types, lighting

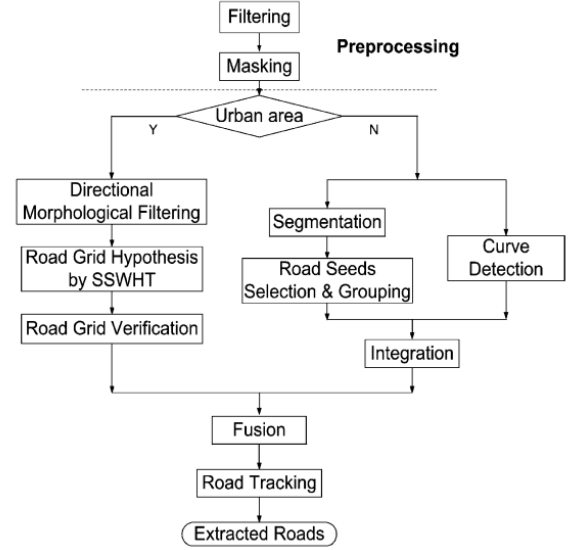


Fig. 1. Processing flow chart for the integrated road mapping system.

Fig. 1. Road extraction workflow [1]

conditions, or varying levels of vegetation. As a result, heuristic approaches are generally considered inferior [10] to modern data-driven techniques for high-variability environments.

In classical methods, roads are often first identified using shape and homogeneity cues. For curvilinear suburban roads, seeds can be selected from homogeneous segments and grouped using perceptual grouping theory. Additional structural cues are then extracted using multi-scale curvilinear detectors based on differential geometry [1].

A. Preprocessing

Typical preprocessing steps include:

- Morphological opening and closing operations [1]
- Median filtering with a 7×7 kernel [1]
- Masking using the Normalized Difference Vegetation Index (NDVI) [1]
- Edge detection using the Nevatia–Babu operator, which provides sharper results than Sobel filters [1]

The preprocessing methods mentioned above are just a small quantity of the vast plethora of possibilities, regurally sharing the same idea, boil down the original image to the most crude and important features. After having said features it is possible to process the image in order to extrac relevant results in this case road location.

B. Road Extraction Workflow

Hough transforms are frequently employed for detecting straight-line segments in road extraction tasks. The standard Hough transform operates on binary images; however, several extensions have been proposed to improve robustness. One such extension is the Spatial Signature Weighted Hough Transform (SSWHT), which incorporates local intensity signatures to enhance detection performance. This technique is

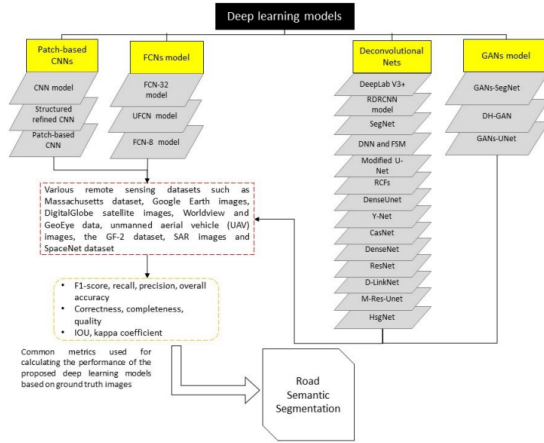


Figure 2. Road semantic segmentation using different deep learning models from remote sensing datasets.

Fig. 2. Road extraction using data-driven techniques [10]

the primary method adopted in the reviewed study. Although alternative approaches exist, they are not examined in detail in this survey. Focusing on the selected work, the complete road extraction workflow is illustrated in fig. 1.

Despite achieving an average performance of 86.0%, as shown in table I, this method suffers from the same limitations as other heuristic approaches when applied to complex or heterogeneous terrain. These shortcomings are regarded as a reason for the transition of both industry and academic research toward data-driven methods.

IV. DATA-DRIVEN METHODS

Recent research increasingly favors data-driven approaches based on deep learning. Unlike heuristic methods, these approaches typically require little to no explicit pre-processing, as feature extraction is learned directly by the model during training. Data-driven methods can vary widely in architectural design; however, this section focuses on the four most prominent architectural families identified in literature and describes their core characteristics and typical applications. It should be noted that data-driven methods typically require substantially larger amounts of training data such as high-quality satellite imagery to be trained effectively.

A. Patch-Based CNN Models

Much like Heuristic methods Convolutional Neural Nets (CNN) also have a preprocessing feature extraction, using convolutional layers, however instead of having set rules to classify the image, these features are sent to a fully connected neural network that classifies the image based on these features, as illustrated in fig. 3.

Patch-based CNN models operate by dividing a large satellite image into smaller, fixed-size patches, which are then processed independently by a convolutional neural network. For each patch, the model predicts the probability of the central pixel or the entire patch belonging to a road class. The final segmentation map is produced by sliding the patch window

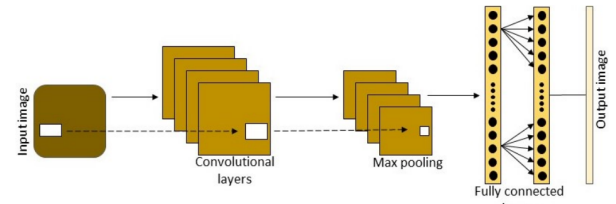


Figure 3. General architecture of the patch-level CNNs model.

Fig. 3. General architecture of the patch-level CNNs model. architecture of the patch-level CNNs model. [10]

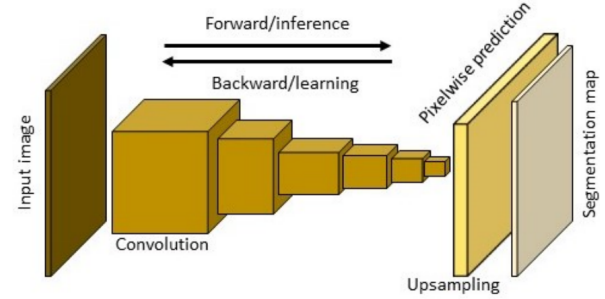


Figure 4. General architecture of FCNs model.

Fig. 4. General architecture of FCNs model [10]

across the image with a predefined stride and assembling the individual predictions into a full-sized label map. This approach allows the network to focus on local spatial context and reduces memory requirements during training. However, because each patch is processed independently, patch-based methods may suffer from spatial inconsistencies and fragmented road segments, particularly at patch boundaries.

B. FCN Models

Fully Convolutional Networks (FCNs) replace the fully connected neural net layers of traditional CNNs with not fully connected ones, enabling dense, pixel-wise prediction over the entire image in a single forward pass, as shown in fig. 4. By operating on full-resolution images, FCNs preserve spatial relationships and improve global consistency compared to patch-based methods. FCNs typically employ downsampling through pooling layers to capture high-level semantic features, followed by upsampling to recover spatial resolution. Skip connections are often used to combine coarse semantic information with fine-grained spatial details. Despite these improvements, FCNs may still produce broken or incomplete road segments, particularly in regions with occlusions, shadows, or highly curved roads.

C. DCED Models

Deep convolutional encoder-decoder (DCED) architectures consist of an encoder that progressively reduces spatial resolution while learning abstract feature representations, just like a CNN or FCN, followed by a decoder that reconstructs the segmentation map at the original image resolution. The

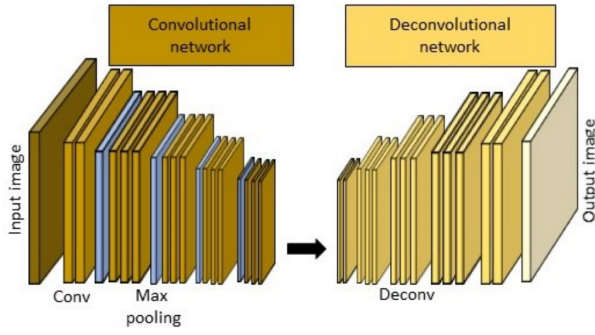


Figure 5. General architecture of deconvolutional networks.

Fig. 5. General architecture of deconvolutional network [10]

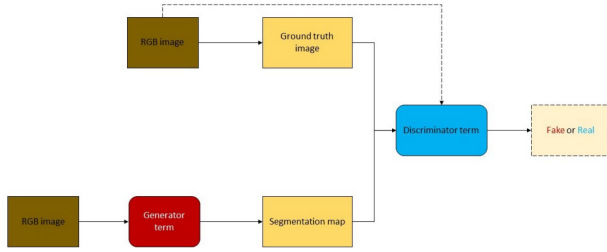


Figure 6. Generic architecture of GANs model.

Fig. 6. Generic architecture of GANs mod [10]

decoder typically uses upsampling or transposed convolutions, often combined with skip connections from corresponding encoder layers to recover spatial detail, as illustrated in fig. 5. These models are well suited for road extraction tasks due to their ability to capture both global context and fine structural details, resulting in smoother and more continuous road predictions. However, DCED models generally require higher computational resources and memory, making them more expensive to train and deploy compared to simpler CNN or FCN architectures.

D. GAN-Based Models

Generative Adversarial Network (GAN) based approaches frame road extraction as an image-to-image translation problem. A generator network produces a road segmentation map from the input image, while a discriminator network attempts to distinguish between ground-truth labels and generated outputs. Through adversarial training, the generator learns to produce more realistic and spatially coherent segmentation results, as depicted in fig. 6. GAN-based models are particularly effective at sharpening road boundaries and enforcing structural consistency. However, they are often more difficult to train due to instability and convergence issues, especially when applied to large-scale or highly diverse datasets. Careful tuning of training strategies and loss functions is typically required to achieve stable performance.

V. RESULTS

With a clear understanding of how each architecture operates, the next section presents the results of the 20 surveyed

papers, ranked according to the average of the reported evaluation metrics. It should be noted that the values shown correspond to the lowest performance reported in each study. This approach was chosen because researchers typically publish their best-performing results, which may not fully represent the typical or average performance of the method. By focusing on the lower-end outcomes, we aim to provide a more realistic comparison of each approach's robustness and reliability.

TABLE I
DATA RESULTS OF EACH METHOD - PART 1

Method	Paper	Model	Avg	F1 Score (1)	mIoU (2)	Completeness (3)	Correctness (4)	Quality (5)	Recall (6)	IoU (7)	precision (8)	accuracy (9)
Heuristic	[1]	Road tracking	86.0%	-	-	86%	92%	80%	-	-	-	-
	[1]	Fusion of two detectors	84.33%	-	-	84%	91%	78%	-	-	-	-
	[1]	Detector I	82.33%	-	-	81%	91%	75%	-	-	-	-
CNN	[15]	SEEDS-MCNN	82.2%	79%	-	-	-	-	80.4%	-	78%	92.2%
	[7]	ResNet-24+	81%	81%	-	-	-	-	-	-	-	-
	[12]	Rasha et al.	80.9%	-	-	-	80.9%	-	-	-	-	-
	[20]	1D-CNN	79.72%	73.4%	-	-	-	-	99.3%	-	58.2%	88%
	[15]	SLIC-CNN	77.95%	72.1%	-	-	-	-	96.2%	-	58.1%	89.2%
	[15]	Pixel-CNN	77.72%	71.2%	-	-	-	-	96.2%	-	57%	86.5%
	[20]	2D-CNN	72.15%	66.6%	-	-	-	-	72.7%	-	61.4%	87.9%
	[7]	ResNet-24	69.5%	81%	58%	-	-	-	-	-	-	-
	[8]	RSRCNN	40.38%	20.8%	-	-	-	-	61.3%	-	12.5%	70.9%
FCN	[6]	FCN-32	86.23%	-	-	74.2%	98.7%	85.8%	-	-	-	-
	[5]	Dice+BCE	80.11%	84.41%	-	-	82.91%	-	-	73.02%	-	-
	[5]	WBCE	78.99%	83.64%	-	-	81.43%	-	-	71.89%	-	-
	[6]	RSRCNN + road struct weight based	73.03%	66.2%	-	-	-	-	72.9%	-	60.6%	92.4%
	[19]	DeepLabv3+	69.28%	72.39%	-	-	-	56.4%	62.67%	-	85.67%	-
	[17]	DeepLabv3+	40.91%	-	-	-	-	-	-	40.91%	-	-
	[11]	Enlarged FCN-4s	63%	-	52%	-	-	-	66%	-	71%	-
	[17]	FCN-8	54.73%	-	-	-	-	-	66.56%	42.92%	54.72%	-
	[11]	Pruned FCN-2s	53.67%	-	42%	-	-	-	61%	-	58%	-
	[11]	FCN-8s	40.67%	-	30%	-	-	-	49%	-	43%	-

TABLE II
DATA RESULTS OF EACH METHOD - PART 2

Method	Paper	Model	Avg	F1 Score (1)	mIoU (2)	Completeness (3)	Correctness (4)	Quality (5)	Recall (6)	IoU (7)	precision (8)	accuracy (9)
DCED	[20]	UFCN	97.25%	96.7%	-	-	-	-	96.1%	-	97.3%	98.9%
	[15]	Road-RCF	93.02%	91.5%	-	-	-	-	98.5%	-	85.8%	96.3%
	[21]	CasNet	91.72%	-	-	-	-	88.84%	92.14%	-	94.18%	-
	[15]	UNet	90.03%	88.9%	-	-	-	-	88.5%	-	89.3%	93.4%
	[13]	ELU-SegNet-RL	85.73%	85.7%	-	-	-	-	86.1%	-	85.4%	-
	[13]	ELU-SegNet-R	81.3%	81.2%	-	-	-	-	84.7%	-	78%	-
	[5]	Dice	80.52%	84.97%	-	-	83.63%	-	-	73.87%	-	-
	[14]	Modified UNet	80.33%	74.54%	-	-	-	-	75.48%	-	74.15%	97.14%
	[13]	ELU-SegNet	79.1%	78.8%	-	-	-	-	73.3%	-	85.2%	-
	[5]	Softmax	78.79%	83.55%	-	-	81.08%	-	-	71.75%	-	-
	[16]	DenseUNet	74.3%	74.07%	-	-	-	-	70.41%	74.47%	78.25%	-
	[16]	GL-Dense-UNet	73.82%	73.98%	-	-	-	-	70.09%	72.73%	78.48%	-
	[16]	FRRN-B	70.57%	66.71%	-	-	-	-	64.87%	74.22%	76.51%	-
	[19]	UNet	68.56%	72.10%	-	-	-	55.64%	64.04%	-	82.47%	-
	[16]	SegNet	66.08%	62.64%	-	-	-	-	71.33%	71.91%	61.35%	-
	[16]	UNet	65.36%	60.78%	-	-	-	-	70.81%	70.91%	58.92%	-
	[7]	UNet	57.5%	72%	43%	-	-	-	-	-	-	-
	[17]	Deep Res. UNet	27.31%	-	-	-	-	-	-	27.31%	-	-
GAN	[6]	pix2pix D:ResNET	87.63%	-	-	76.3%	100%	86.6%	-	-	-	-
	[18]	DH-GAN D:PathGAN	84.05%	84.05%	-	-	-	-	-	-	-	-
	[18]	DH-GAN + SBO G:UNet D:PathGAN	81.74%	81.74%	-	-	-	-	-	-	-	-
	[9]	GAN	84.28%	-	-	-	-	80.92%	89.81%	-	82.12%	-
	[18]	DH-GAN + Greedy G:UNet D:PathGAN	80.81%	80.81%	-	-	-	-	-	-	-	-
Other	[4]	DNN + FSM	74.67%	-	-	78%	76%	70%	-	-	-	-
	[19]	FPN	71.6%	75.33%	-	-	-	59.75%	70.66%	-	80.67%	-
	[20]	SVM (ML)	66.98%	56.6%	-	-	-	-	90.4%	-	41.1%	76.8%

VI. ERROR MEASUREMENT METHODS

Common evaluation metrics include:

A. F1 Score

$$\frac{1}{N} \sum_{i=0}^N \frac{2 \times \text{Area of overlap}}{\text{Total Area}} \quad (1)$$

B. Mean Intersection over Union (mIoU)

$$\frac{1}{N} \sum_{i=0}^N \frac{\text{Predicted Road} \cap \text{Known Road}}{\text{Predicted Road} \cup \text{Known Road}} \quad (2)$$

C. Completeness

$$\frac{\text{length of matched reference}}{\text{length of reference}} \quad (3)$$

D. Correctness

$$\frac{\text{length of matched extraction}}{\text{length of extraction}} \quad (4)$$

E. Quality / Q

$$\frac{\text{length of matched extraction}}{\text{length of extraction} + \text{length of unmatched reference}} \quad (5)$$

F. Recall / COM

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (6)$$

G. IoU

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives} + \text{True Negatives}} \quad (7)$$

H. precision / COR

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (8)$$

I. accuracy

$$\frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{False Positives} + \text{True Negatives} + \text{False Negatives}} \quad (9)$$

VII. CONCLUSION AND COMMENTS

As shown in table I and table II, as well as section VI there are many different ways to measure the efficacy of the many different solutions to this problem. However, as can also be seen in table I and table II different authors opt to use different measurement techniques, this makes comparing different authors' results quite difficult. This problem is exacerbated by the complexity of these solutions, often containing many different parameters that can be tweaked to improve their efficacy, parameters that are seldom shared or even mentioned, making comparing even the same solution implemented by different authors quite frivolous.

For a more in depth analysis of these solutions and the models they propose, a standard efficacy measure toolkit should be developed and it's use incentivised by many different authors.

Although, despite these issues it is still possible to observe some trends within the collected data, these trends hopefully reveal larger solution comparisons. It is important to note that due to a lack of comparable data the conclusions mentioned bellow are at best extrapolations and at worst pure conjecture.

From the gathered data it is possible to see that, well implemented CNN based solutions tend to perform about as well as FCN based solutions and worst than DCED or GAN based solutions. Even still, FCN based solutions have the capacity to outperform CNN based solutions.

Finally, we observe substantial variability in reported results for identical or closely related architectures across different studies. This inconsistency is likely attributable to differences in implementation choices, training procedures, datasets, and evaluation protocols. In many cases, authors may present comparatively weaker baseline architectures in order to highlight the effectiveness of their proposed methods, rather than demonstrating the full potential performance achievable with the same architectural family. As a result, direct performance comparisons between studies should be interpreted with caution, as they may not accurately reflect the true capabilities of the underlying architectures. Making us once again vouch for a creation of a standardized dataset, architecture creation process and training procedure, ensuring a more reliable way to compare models and results analithically. Until this procedure is in place the use of the Massachusetts dataset is recommended, being, to the researchers, knowledge the most widely adopted dataset.

REFERENCES

- [1] X. Jin and C. H. Davis, "An integrated system for automatic road mapping from high-resolution multi-spectral satellite imagery by information fusion," *Information Fusion*, vol. 6, no. 4, pp. 257–273, 2005, fusion of Remotely Sensed Data over Urban Areas. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1566253504000454>
- [2] "Digitalglobe completes quickbird satellite orbit raise," <https://web.archive.org/web/20140714153612/http://media.digitalglobe.com/manual-releases/DigitalGlobe-Completes-QuickBird-Satellite-Orbit-R>, accessed: 2025-11-14.
- [3] "Quickbird wiki article," <https://en.wikipedia.org/wiki/QuickBird>, accessed: 2025-11-14.

- [4] J. Wang, J. Song, M. Chen, and Z. Yang, "Road network extraction: a neural-dynamic framework based on deep learning and a finite state machine," *International Journal of Remote Sensing*, vol. 36, no. 12, pp. 3144–3169, 2015. [Online]. Available: <https://doi.org/10.1080/01431161.2015.1054049>
- [5] Y. Lin, D. Xu, N. Wang, Z. Shi, and Q. Chen, "Road extraction from very-high-resolution remote sensing images via a nested se-deeplab model," *Remote Sensing*, vol. 12, no. 18, 2020. [Online]. Available: <https://www.mdpi.com/2072-4292/12/18/2985>
- [6] N. Varia, A. Dokania, and J. Senthilnath, "Deepext: A convolution neural network for road extraction using rgb images captured by uav," in *2018 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2018, pp. 1890–1895.
- [7] S. Sloan, R. R. Talkhani, T. Huang, J. Engert, and W. F. Laurance, "Mapping remote roads using artificial intelligence and satellite imagery," *Remote Sensing*, vol. 16, no. 5, 2024. [Online]. Available: <https://www.mdpi.com/2072-4292/16/5/839>
- [8] Y. Wei, Z. Wang, and M. Xu, "Road structure refined cnn for road extraction in aerial image," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 5, pp. 709–713, 2017.
- [9] Q. Shi, X. Liu, and X. Li, "Road detection from remote sensing images by generative adversarial networks," *IEEE Access*, vol. 6, pp. 25 486–25 494, 2018.
- [10] A. Abdollahi, B. Pradhan, N. Shukla, S. Chakraborty, and A. Alamri, "Deep learning approaches applied to remote sensing datasets for road extraction: A state-of-the-art review," *Remote Sensing*, vol. 12, no. 9, 2020. [Online]. Available: <https://www.mdpi.com/2072-4292/12/9/1444>
- [11] Z. Zhong, J. Li, W. Cui, and H. Jiang, "Fully convolutional networks for building and road extraction: Preliminary results," in *2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2016, pp. 1591–1594.
- [12] R. Alshehhi, P. R. Marpu, W. L. Woon, and M. D. Mura, "Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 130, pp. 139–149, 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0924271617300096>
- [13] T. Panboonyuen, P. Vateekul, K. Jitkajornwanich, and S. Lawawirojwong, "An enhanced deep convolutional encoder-decoder network for road segmentation on aerial imagery," in *Recent Advances in Information and Communication Technology 2017*, P. Meesad, S. Sodsee, and H. Unger, Eds. Cham: Springer International Publishing, 2018, pp. 191–201.
- [14] A. Constantin, J.-J. Ding, and Y.-C. Lee, "Accurate road detection from satellite images using modified u-net," in *2018 IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)*, 2018, pp. 423–426.
- [15] Z. Hong, D. Ming, K. Zhou, Y. Guo, and T. Lu, "Road extraction from a high spatial resolution remote sensing image based on richer convolutional features," *IEEE Access*, vol. 6, pp. 46 988–47 000, 2018.
- [16] J. Xin, X. Zhang, Z. Zhang, and W. Fang, "Road extraction of high-resolution remote sensing images derived from denseunet," *Remote Sensing*, vol. 11, no. 21, 2019. [Online]. Available: <https://www.mdpi.com/2072-4292/11/21/2499>
- [17] C. Henry, S. M. Azimi, and N. Merkle, "Road segmentation in sar satellite images with deep fully convolutional neural networks," *IEEE Geoscience and Remote Sensing Letters*, vol. 15, no. 12, pp. 1867–1871, 2018.
- [18] D. Costea, A. Marcu, M. Leordeanu, and E. Slusanschi, "Creating roadmaps in aerial images with generative adversarial networks and smoothing-based optimization," in *2017 IEEE International Conference on Computer Vision Workshops (ICCVW)*, 2017, pp. 2100–2109.
- [19] M. Lourenço, D. Estima, H. Oliveira, L. Oliveira, and A. Mora, "Automatic rural road centerline detection and extraction from aerial images for a forest fire decision support system," *Remote Sensing*, vol. 15, no. 1, 2023. [Online]. Available: <https://www.mdpi.com/2072-4292/15/1/271>
- [20] R. Kestur, S. Farooq, R. Abdal, E. Mehraj, O. S. Narasipura, and M. Mudigere, "UFCN: a fully convolutional neural network for road extraction in RGB imagery acquired by remote sensing from an unmanned aerial vehicle," *Journal of Applied Remote Sensing*, vol. 12, no. 1, p. 016020, 2018. [Online]. Available: <https://doi.org/10.1117/1.JRS.12.016020>
- [21] G. Cheng, Y. Wang, S. Xu, H. Wang, S. Xiang, and C. Pan, "Automatic road detection and centerline extraction via cascaded end-to-end convolutional neural network," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 6, pp. 3322–3337, 2017.